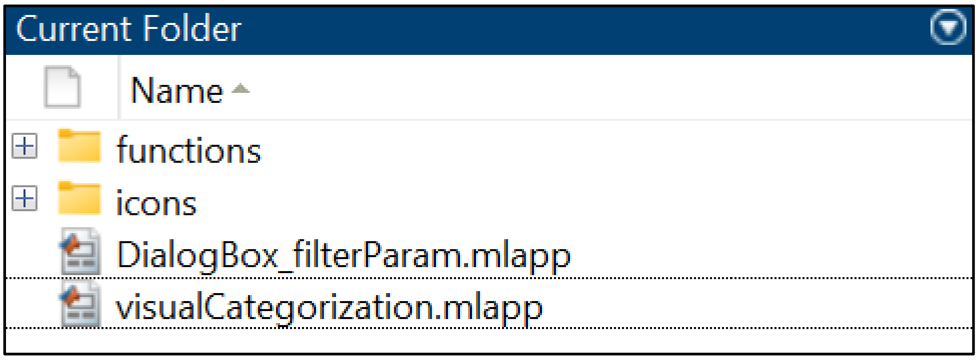
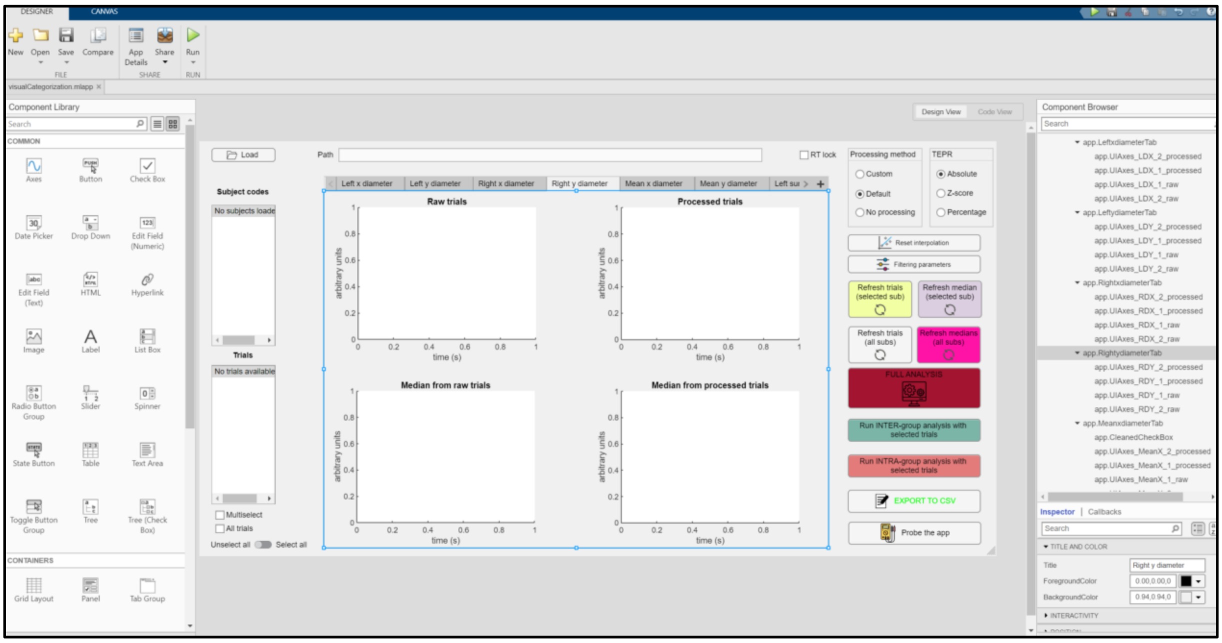
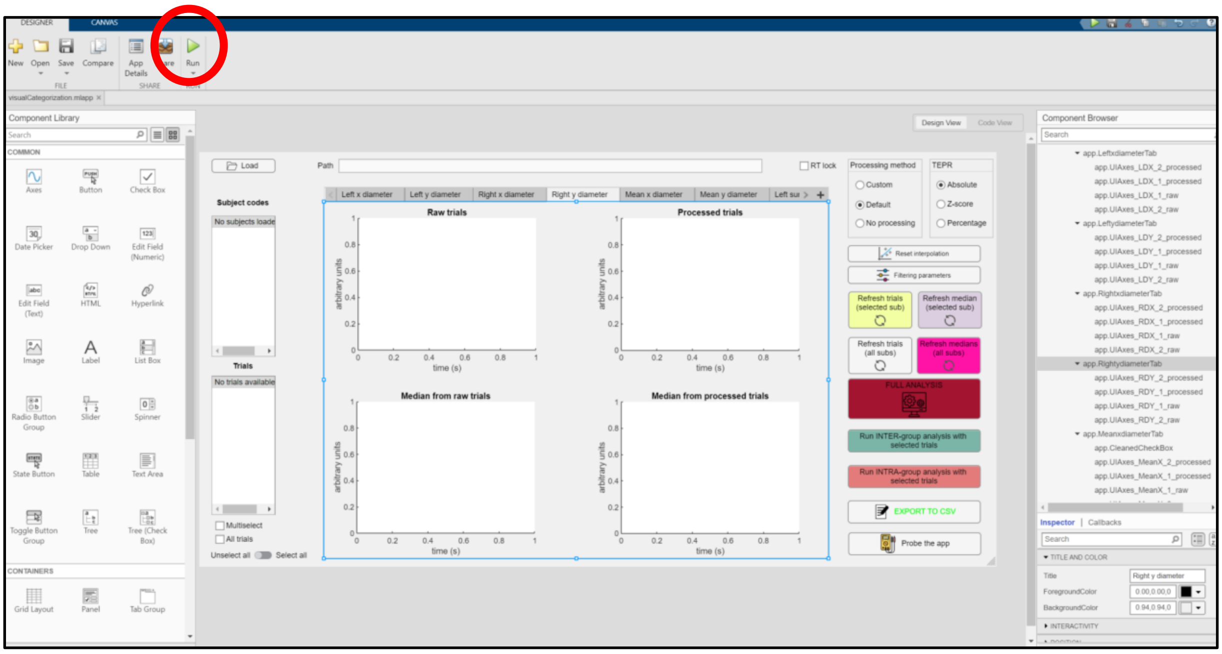
1/ Open **visualCategorization.mlapp** in **code/**



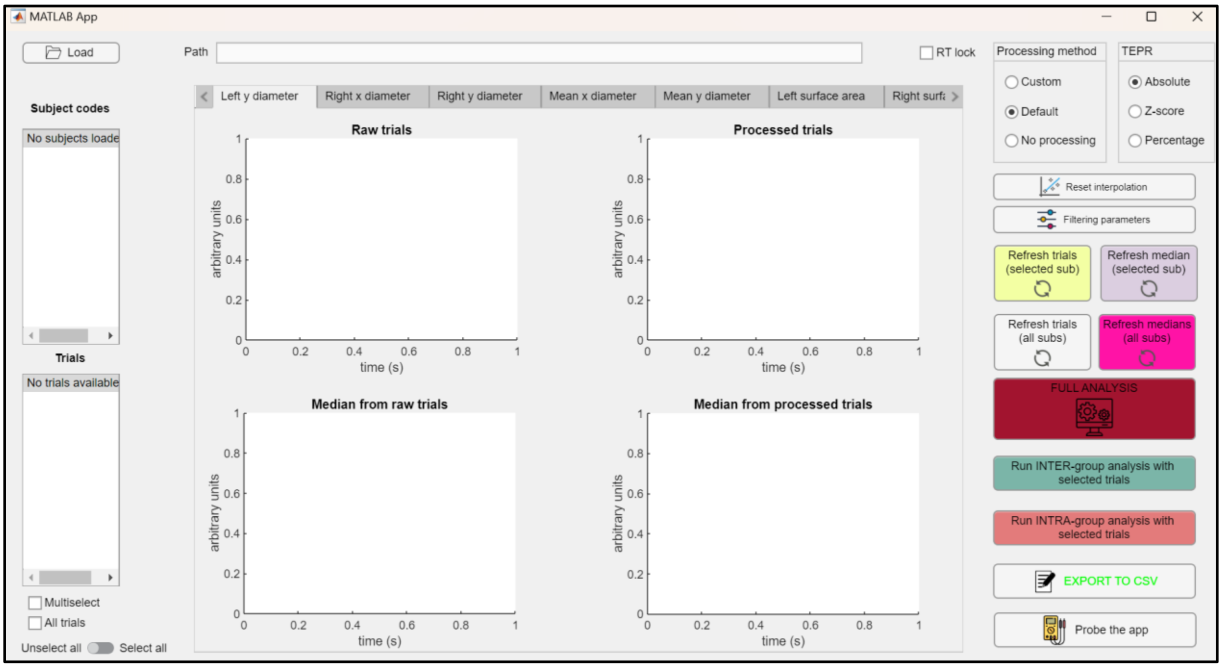
2/ Wait for Matlab App Designer to open with this window



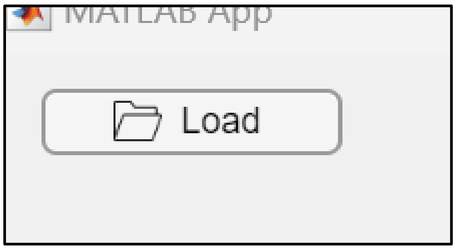
3/ Click on **Run**



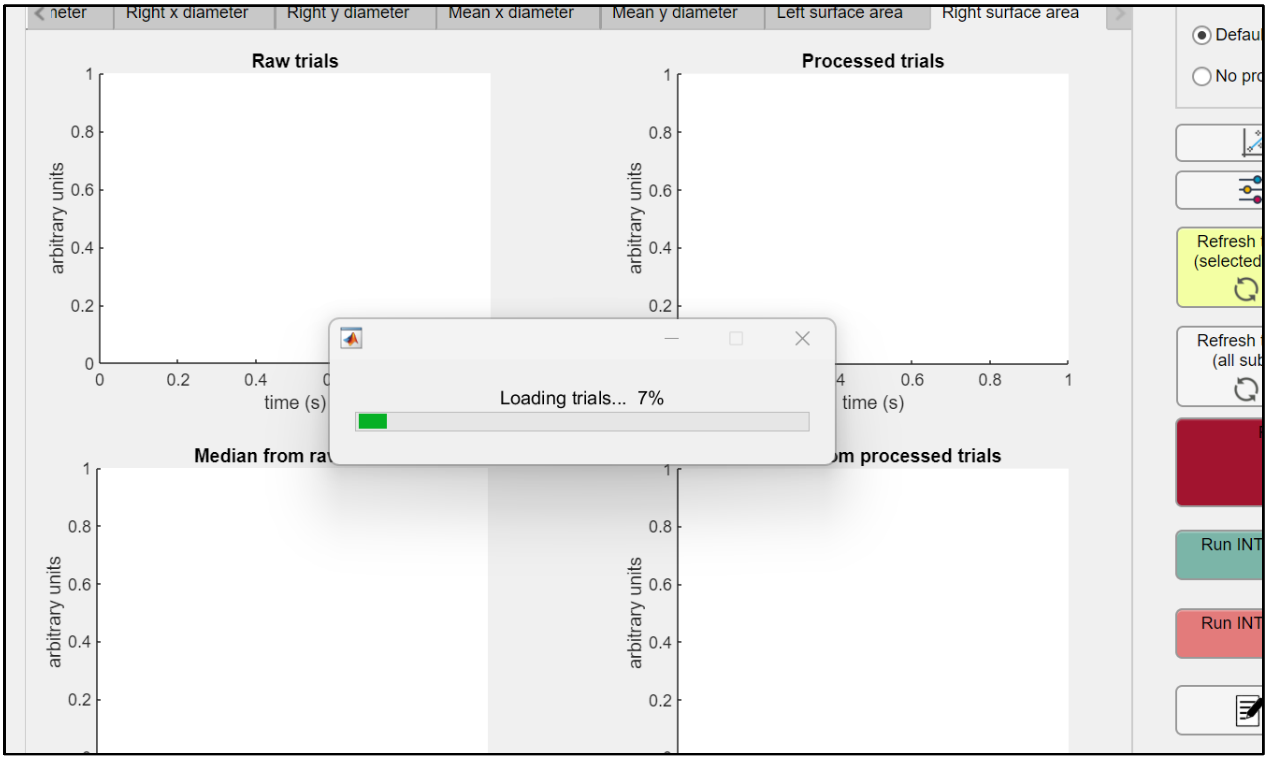
4/ This window opens: it is the **GUI**



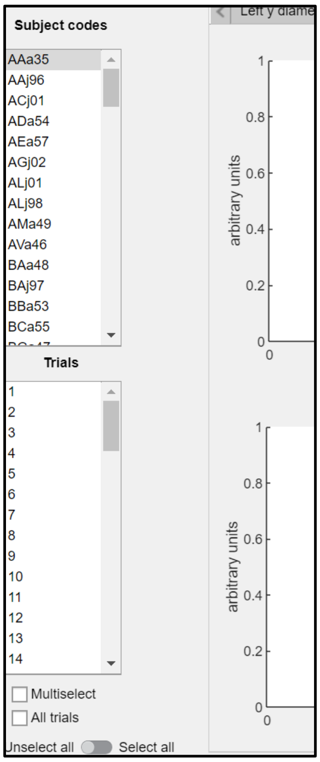
5/ Click on **load** in the top left corner and select **data/**



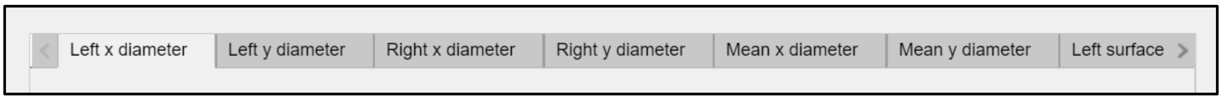
6/ Wait until all trials are loaded



7/ Once the participants have been loaded, they will appear in the "Subject codes" sidebar.

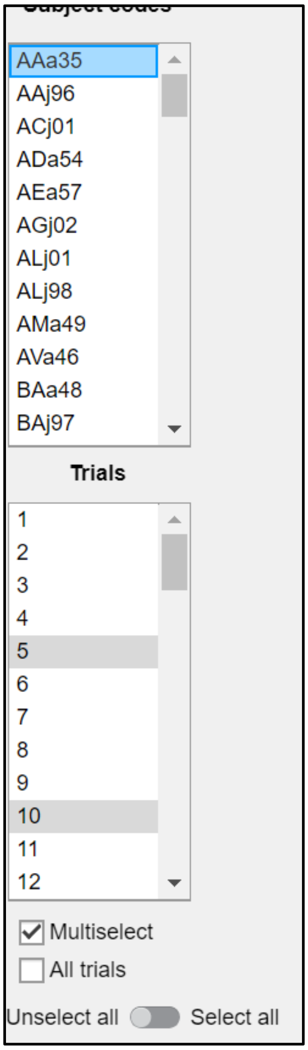


8/ You can choose which diameter to display by clicking on the corresponding panel at the top: the horizontal diameter of the left eye (i.e**. Left x diameter**), the vertical diameter of the left eye (**Left y diameter**), the horizontal diameter of the right eye, the vertical diameter of the right eye, the mean horizontal diameter, the mean vertical diameter, the left pupil area, and the right pupil area. We chose to work with the mean horizontal pupil diameter (**Mean x diameter)** for this study, but you can explore the other measurements as well. (We did not look at the other measurements).



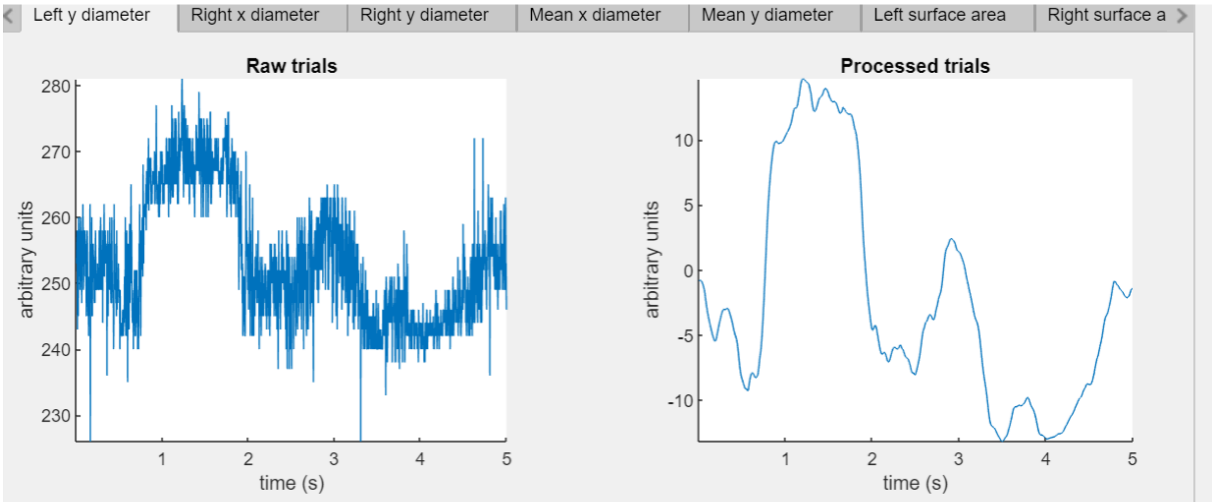
9/ By clicking on **Mean x diameter**, you need to check the **Cleaned** box in the center. There are two options: either you take the average of the two raw signals and then process this average (with the **Cleaned** box **unchecked**), or you process both signals first and then compute their average (with the **Cleaned** box **checked**).

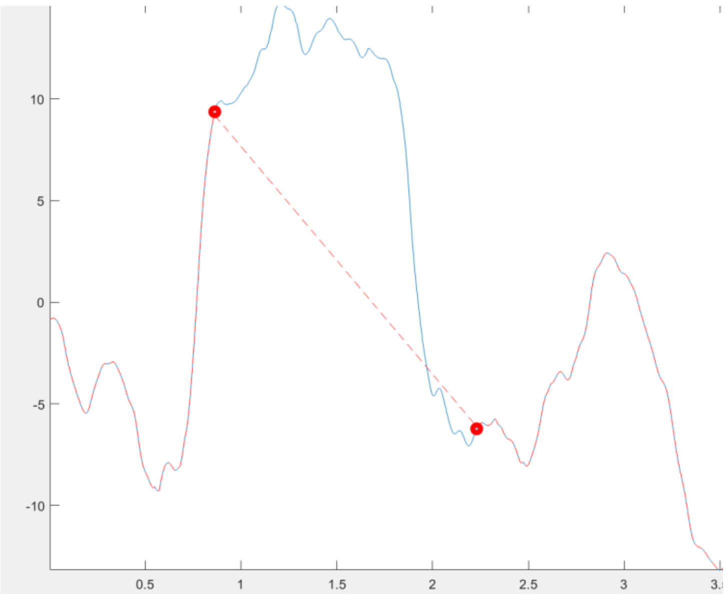
10/ You can then choose the **subjects** and **trials** you want. To display **multiple** trials and participants at once, you first need to check **Multiselect**. If you click **Select all** at the bottom left, all trials from all participants will be selected (the selected trials appear greyed out in the “Trials” section on the left).



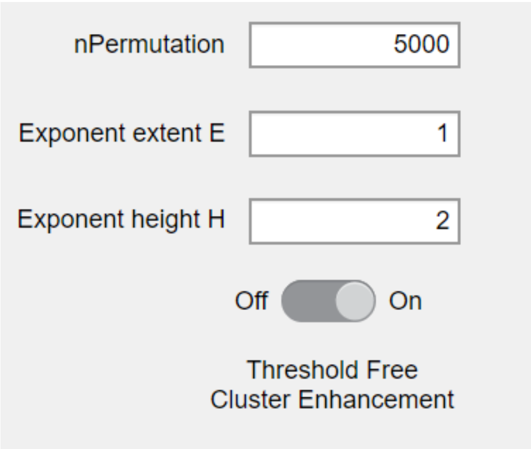
11/ The graph at the **top left** shows the **raw trials**, the one at the **top right** shows the **preprocessed trials** using the "Default" method selected at the top right, the **bottom left** graph displays the **medians** of the **raw trials** for each participant, and the **bottom right** graph shows the medians of the processed trials (green: young participants; blue: older participants; red: average).

12/ You can review each trial and correct it manually if necessary, by interpolating with the help of the mouse. To do this, you need to select the participant and their trial (remember to uncheck **Multiselect** if it's already checked), **and click on the preprocessed trial window** (top right). A new window will then open, and you need to select the portion of the signal to interpolate linearly. To select multiple trials again, remember to recheck **Multiselect**.





13/ Once the participants and their trials have been selected (either one by one or using the **Select all** feature), you need to click **Full Analysis** to start the within-subject and between-subject statistical analysis (Threshold Free Cluster Enhancement, TFCE). The default number of permutations is 5 000, which allows the analyses to be completed within a few days. To change this number, as well as the processing or statistical analysis parameters, click **Filtering parameters**, adjust the settings, and click **OK**. You can choose to run only the between-group analysis (**Run INTER-group analysis with selected trials**) or only the within-group analysis (**Run INTRA-group analysis with selected trials**). By default, TFCE is selected, but you can switch to Cluster-Based Permutation Testing (CBPT) by turning TFCE Off.



14/ Once the analysis is complete, the graphs will be generated one by one, and a message in the MatLab console will indicate when it's finished and how long the script took to run.

15/ Click **EXPORT TO CSV** to export the following files into **outputFiles/** :

- **categorisationResult** – contains the pupillary metrics and behavioral data for each participant (for subject-level analyses, e.g., correlations, multiple regressions, linear models, etc.).
- **trialWiseGLMM** – contains the pupillary metrics and behavioral data for each trial of each participant (for trial-level analyses, e.g., mixed linear models with the participant as a random effect).
- **trialValidity** – contains the number of valid trials for each condition and participant, as well as the number of subjects performing above chance on the recognition task.

PS: Matlab's App Designer isn't very optimized, so the application's display may vary depending on the screen, and you shouldn't click buttons too quickly in succession. If you encounter any issues, please contact me by email**: adrian.ruizchiapello@gmail.com**